

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 40%

Annexure A

SHORT ANSWER TEST

Code of Conduct:

I P Surendra Kumar Reddy (192425224) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: P Surendra Kumar Reddy

Short Answer Test

1. A company needs to transfer a **150 MB** design file from a client computer to a remote server over a network with a bandwidth of **50 Mbps**. The Transport Layer divides the file into **1500-byte segments**, while the Data Link Layer encapsulates each segment into frames by adding **40 bytes of header and trailer**. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer, and estimate the total transmission time (ignoring propagation delay). Explain how the Transport and Data Link layers ensure reliable communication.

Given

- File size = **150 MB**
- Bandwidth = **50 Mbps**
- Segment size = **1500 bytes**
- Data Link overhead = **40 bytes/frame**

Step 1: Convert file size

$$150 \text{ MB} = 150 \times 10^6 = 150,000,000 \text{ bytes}$$

Step 2: Number of segments

$$\frac{150,000,000}{1500} = 100,000 \text{ segments}$$

Since one segment becomes one frame,

$$\text{Frames} = 100,000$$

Step 3: Total overhead

Overhead per frame = 40 bytes

$$100,000 \times 40 = 4,000,000 \text{ bytes}$$

$$= 4 \text{ MB}$$

Step 4: Transmission time

Total transmitted data

$$150 + 4 = 154 \text{ MB}$$

Convert into bits

$$154 \times 8 = 1232 \text{ Mb}$$

Bandwidth

$$50 \text{ Mbps}$$

Time

$$\frac{1232}{50} = 24.64 \text{ seconds}$$

Answer

- Number of segments = **100,000**
- Number of frames = **100,000**
- Total overhead = **4 MB**
- Transmission time $\approx$ **24.64 seconds**

Explanation

Transport Layer

- Divides the file into segments.
- Performs sequencing, acknowledgments, flow control, and error recovery.

Data Link Layer

- Encapsulates segments into frames.
- Detects transmission errors using CRC.
- Ensures reliable communication between adjacent devices.

2.A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions.

Explain how flow control improves communication efficiency.

Given

- Sliding window size = **6**
- Total frames = **48**
- Frame size = **1024 bytes**

Step 1: Number of windows

$$\frac{48}{6} = 8$$

Windows required = **8**

Step 2: Retransmissions

One frame lost in every window

$$8 \times 1 = 8$$

Retransmissions = **8**

Answer

- Windows required = **8**
- Retransmitted frames = **8**

Flow Control

Flow control prevents the sender from overwhelming the receiver. It regulates the transmission rate, minimizes packet loss, avoids congestion, improves throughput, and ensures efficient communication.

3.A **12,000-byte** IP packet is transmitted through a router where the Maximum Transmission Unit (MTU) is **1500 bytes**. Assuming each fragment requires a **20byte IP header**, calculate the total number of fragments created and the total header overhead introduced. Explain how the Network Layer ensures successful delivery of fragmented packets to the destination.

Given

- IP packet = **12,000 bytes**
- MTU = **1500 bytes**
- IP header = **20 bytes**

Step 1: Payload per fragment

$$1500 - 20 = 1480 \text{ bytes}$$

Step 2: Number of fragments

$$\frac{12000}{1480} = 8.1$$

Hence,

Fragments = **9**

Step 3: Header overhead

Each fragment has a 20-byte header.

$$9 \times 20 = 180 \text{ bytes}$$

Answer

- Number of fragments = **9**
- Total header overhead = **180 bytes**

Network Layer Explanation

The Network Layer performs fragmentation when packets exceed the MTU. Each fragment contains its own IP header with identification, fragment offset, and more-

fragments flag. At the destination, all fragments are reassembled into the original packet, ensuring successful delivery across networks.

4. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions.

Explain how flow control improves communication efficiency.

Given

- Sliding window size = **6 frames**
- Total frames = **48**
- Frame size = **1024 bytes**

Step 1: Number of transmission windows

$$\frac{48}{6} = 8$$

Transmission windows = 8

Step 2: Retransmissions

One frame is lost in every window.

$$8 \times 1 = 8$$

Retransmissions = 8 frames

Final Answer

- Transmission windows = **8**
- Retransmissions = **8 frames**

Flow Control Explanation

Flow control regulates the sender's transmission speed according to the receiver's capacity. It prevents buffer overflow, minimizes congestion, reduces

retransmissions, improves throughput, and ensures efficient and reliable communication.

5.A multimedia application transfers a **600 MB** video file. The Session Layer inserts a synchronization checkpoint after every **60 MB** of transmitted data. During transmission, a failure occurs after **390 MB** has been transferred. Determine the number of checkpoints created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the Session Layer.

Given

- Video file = **600 MB**
- Checkpoint after every **60 MB**
- Failure after **390 MB**

Step 1: Number of checkpoints before failure

$$\frac{390}{60} = 6.5$$

Completed checkpoints:

60, 120, 180, 240, 300, 360

Checkpoints created = 6

Step 2: Data to retransmit

Failure occurs at **390 MB**.

Last checkpoint = **360 MB**

Retransmission:

$$390 - 360 = 30 \text{ MB}$$

Final Answer

- Checkpoints created = **6**
- Data retransmitted = **30 MB**

Session Layer Explanation

The Session Layer establishes, maintains, and terminates communication sessions. Synchronization checkpoints allow interrupted transmissions to resume from the last checkpoint instead of restarting from the beginning, saving time and bandwidth.

6. Before transmission, the Presentation Layer compresses a **900 MB** file by **40%**. The compressed data is encrypted, adding **5%** overhead to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.

Given

- Original file = **900 MB**
- Compression = **40%**
- Encryption overhead = **5%**

Step 1: Compressed file size

40% reduction means 60% remains.

$$900 \times 0.60 = 540 \text{ MB}$$

Step 2: Encryption overhead

$$540 \times 0.05 = 27 \text{ MB}$$

Encrypted size

$$540 + 27 = 567 \text{ MB}$$

Step 3: Percentage reduction

Reduction

$$900 - 567 = 333 \text{ MB}$$

Percentage

$$\frac{333}{900} \times 100 = 37\%$$

Final Answer

- Compressed file size = **540 MB**
- Encrypted file size = **567 MB**
- Total reduction = **37%**

Presentation Layer Explanation

The Presentation Layer performs data compression, encryption, decryption, and format conversion. Compression reduces transmission time and bandwidth usage, while encryption protects data confidentiality and integrity during transmission.

7. An FTP application transfers **3 GB** of data over a network with an effective throughput of **120 Mbps**. The Application Layer divides the data into requests of **3 MB** each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.

Given

- Data = **3 GB**
- Throughput = **120 Mbps**
- Request size = **3 MB**

Step 1: Convert data

$$3 \text{ GB} = 3000 \text{ MB}$$

Step 2: Number of requests

$$\frac{3000}{3} = 1000$$

Requests = 1000

Step 3: Transmission time

Convert data into megabits

$$3000 \times 8 = 24000 \text{ Mb}$$

Time

$$\frac{24000}{120} = 200 \text{ seconds}$$

Final Answer

- Number of requests = **1000**
- Transmission time = **200 seconds**

Application Layer Explanation

The Application Layer provides services such as FTP, HTTP, SMTP, and DNS. It enables file transfer, user authentication, error reporting, and communication between network applications and users.

8.A packet experiences the following processing delays at different OSI layers: Application Layer = **4 ms**, Transport Layer = **3 ms**, Network Layer = **5 ms**, Data Link Layer = **2 ms**, and Physical Layer = **1 ms**. The propagation delay is **18 ms**, and the transmission delay is **12 ms**. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.

Given

Layer	Delay
Application	4 ms
Transport	3 ms
Network	5 ms
Data Link	2 ms
Physical	1 ms
Propagation	18 ms
Transmission	12 ms

Total Delay

$$4 + 3 + 5 + 2 + 1 + 18 + 12 \\ = 45 \text{ ms}$$

Final Answer

End-to-end delay = 45 ms

Layer Contributions

- **Application Layer:** Provides network services to applications.
- **Transport Layer:** Ensures reliable delivery and flow control.
- **Network Layer:** Handles routing and logical addressing.
- **Data Link Layer:** Performs framing and error detection.
- **Physical Layer:** Transmits bits over the communication medium.
- **Propagation Delay:** Time for the signal to travel through the medium.

- **Transmission Delay:** Time required to place all bits onto the transmission medium.

9. A network transfers **80 MB** of useful data using frames of **1600 bytes**, where each frame contains **80 bytes** of protocol overhead added by different OSI layers. Calculate the total number of frames required, the total overhead transmitted, and the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

Given

- Useful data = **80 MB**
- Frame size = **1600 bytes**
- Overhead = **80 bytes/frame**

Step 1: Useful payload

$$1600 - 80 = 1520 \text{ bytes}$$

Step 2: Number of frames

Useful data

$$80 \text{ MB} = 80,000,000 \text{ bytes}$$

$$\frac{80,000,000}{1520} = 52,632 \text{ frames (approx.)}$$

Step 3: Total overhead

$$52,632 \times 80 = 4,210,560 \text{ bytes}$$

≈ 4.21 MB

Step 4: Transmission Efficiency

$$\frac{1520}{1600} \times 100 = 95\%$$

Final Answer

- Frames required $\approx$ **52,632**
- Total overhead $\approx$ **4.21 MB**
- Transmission efficiency = **95%**

Explanation

Protocol overhead consumes additional bandwidth because headers and trailers are transmitted along with user data. Excessive overhead reduces efficiency, while optimized protocols improve network performance.

10.A hospital transfers a patient's medical record of **240 MB** through an IoT enabled healthcare network. The Presentation Layer compresses the data by **30%**, the Transport Layer divides the compressed data into **1400-byte segments**, and the Data Link Layer adds **60 bytes** of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.

Given

- Original file = **240 MB**
- Compression = **30%**
- Segment size = **1400 bytes**
- Frame overhead = **60 bytes**

Step 1: Compressed size

70% remains after compression.

$$240 \times 0.70 = 168 \text{ MB}$$

Step 2: Convert to bytes

168,000,000 bytes

Step 3: Number of segments

$$\frac{168,000,000}{1400} = 120,000$$

Segments = 120,000

Step 4: Protocol overhead

$$120,000 \times 60 = 7,200,000 \text{ bytes}$$

= 7.2 MB

Step 5: Final transmitted data

$$168 + 7.2 = 175.2 \text{ MB}$$

Final Answer

- Compressed file size = **168 MB**
- Segments transmitted = **120,000**
- Protocol overhead = **7.2 MB**
- Total data transmitted = **175.2 MB**

Explanation

The Presentation Layer compresses the data and provides encryption if required. The Transport Layer divides the data into segments for reliable delivery. The Data Link Layer encapsulates each segment into frames by adding headers and trailers for error detection. Finally, the Physical Layer transmits the complete framed data across the network. Together, these OSI layers ensure efficient, secure, and reliable communication.